FloraDoc: AI-Powered Plant Disease Detection

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Outline

1. Problem Statement
2. Introduction
3. Design and Architecture
4. Implementation Details

The Challenge: Agricultural Crop Loss

The Problem

Farmers and home gardeners often struggle with the early and accurate identification of plant diseases. This uncertainty leads to:

- Significant crop loss and economic damage.
- Incorrect application of pesticides, harming the environment.
- Delays in treatment, allowing diseases to spread.



Figure: A leaf showing signs of

Our Goal

To develop a simple, accessible, and instant

Our Solution: A Mobile App for Instant Diagnosis

FloraDoc is a cross-platform mobile application that leverages the power of Artificial Intelligence to identify plant diseases from a single photograph.

Key Features

- **Instant Identification:** Snap a photo of a plant leaf and get a diagnosis in seconds.
- **High Accuracy:** Powered by a deep learning model trained on thousands of images.
- **Treatment Suggestions:** Provides recommendations for organic and chemical treatments.
- **User-Friendly Interface:** Designed for simplicity and ease of use.

How FloraDoc Works: A High-Level Overview

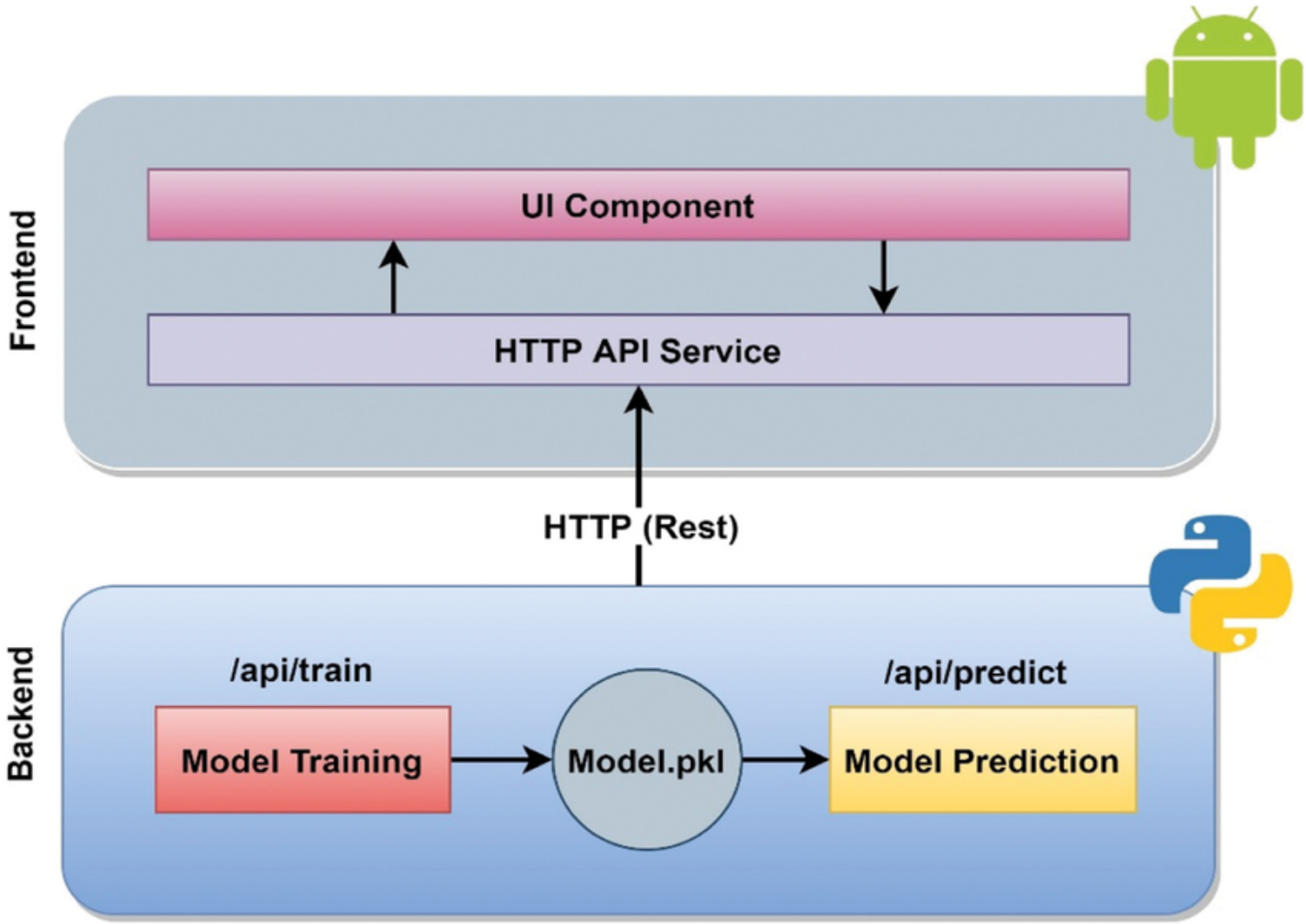


Figure: The three main components: Mobile App, Backend API, and ML Model.

The Tools and Technologies We Used

Frontend (Mobile App)

- **Framework:** Flutter
- **Language:** Dart
- **Key Libraries:** `camera`, `http`

Backend (API Server)

- **Framework:** FastAPI
- **Language:** Python
- **Deployment:** Docker, AWS

Machine Learning Model

- **Framework:** TensorFlow/Keras
- **Architecture:** MobileNetV2 (CNN)
- **Dataset:** PlantVillage Dataset
- **Training:** Transfer Learning

A Look at the Prediction Endpoint

Here is a simplified version of the Python code (using FastAPI) that handles image uploads and returns a prediction.

```
from fastapi import FastAPI, UploadFile, File
from PIL import Image
import tensorflow as tf

app = FastAPI()
model = tf.keras.models.load_model("plant_disease_model.h5")
class_names = ["Healthy", "Rust", "Powdery Mildew"]

@app.post("/predict")
async def predict(file: UploadFile = File(...)):
    # Read and preprocess the image
    image = Image.open(file.file)
    image = image.resize((224, 224))
    img_array = tf.keras.utils.img_to_array(image)
    img_array = tf.expand_dims(img_array, 0) # Create batch

    # Make prediction
    predictions = model.predict(img_array)
```

Conclusion & Questions

Thank You!

Questions?